

# Euclid, Book I, Proposition I.2

## To Place at a Given Point a Straight Line Equal to a Given Straight Line

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This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

### 1 Statement

Given a point  $A$  and a segment  $BC$ , construct a point  $D$  such that

$$AD \cong BC.$$

### 2 Euclid's Classical Construction

Euclid's construction proceeds as follows.

1. Construct the equilateral triangle  $DAB$  on the segment  $AB$ .
2. Produce the side  $DB$  beyond  $B$ .
3. Produce the side  $DA$  beyond  $A$ .
4. Draw the circle centered at  $B$  with radius  $BC$ .
5. Draw the circle centered at  $D$  with radius  $DG$ .
6. The constructed point  $L$  satisfies

$$AL \cong BC.$$

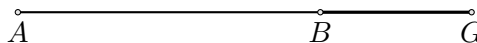
#### Step 1. Given data

The construction starts with a given point  $A$  and a given segment  $BG$ .



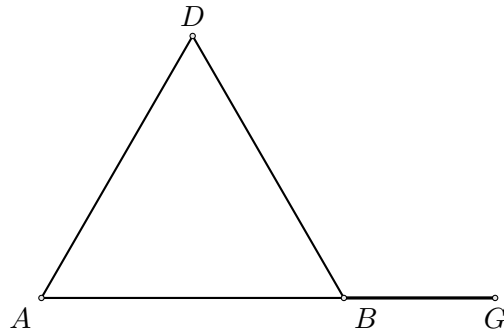
#### Step 2. Join $AB$

Join the given point  $A$  to the endpoint  $B$  of the given segment.



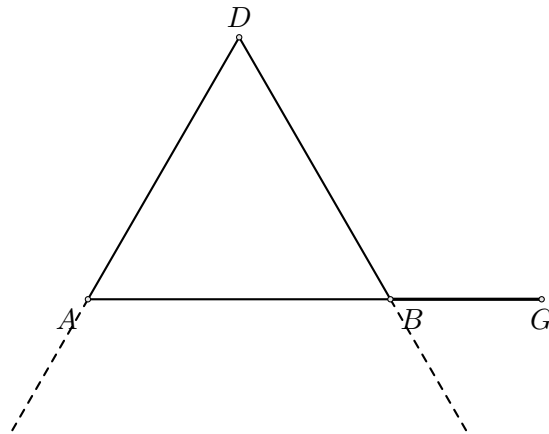
**Step 3. Construct the equilateral triangle  $DAB$**

Construct the equilateral triangle  $DAB$  on the segment  $AB$ .



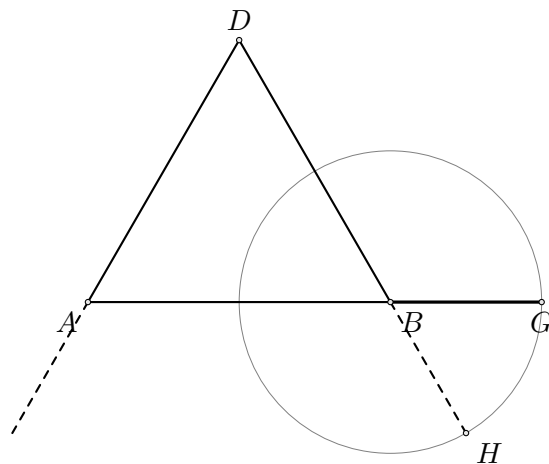
**Step 4. Produce  $DA$  and  $DB$**

Produce  $DA$  beyond  $A$  and  $DB$  beyond  $B$ .



**Step 5. Draw the circle centered at  $B$**

Draw the circle centered at  $B$  through  $G$ . Let  $H$  be the intersection of this circle with the line  $DB$  on the side beyond  $B$ .

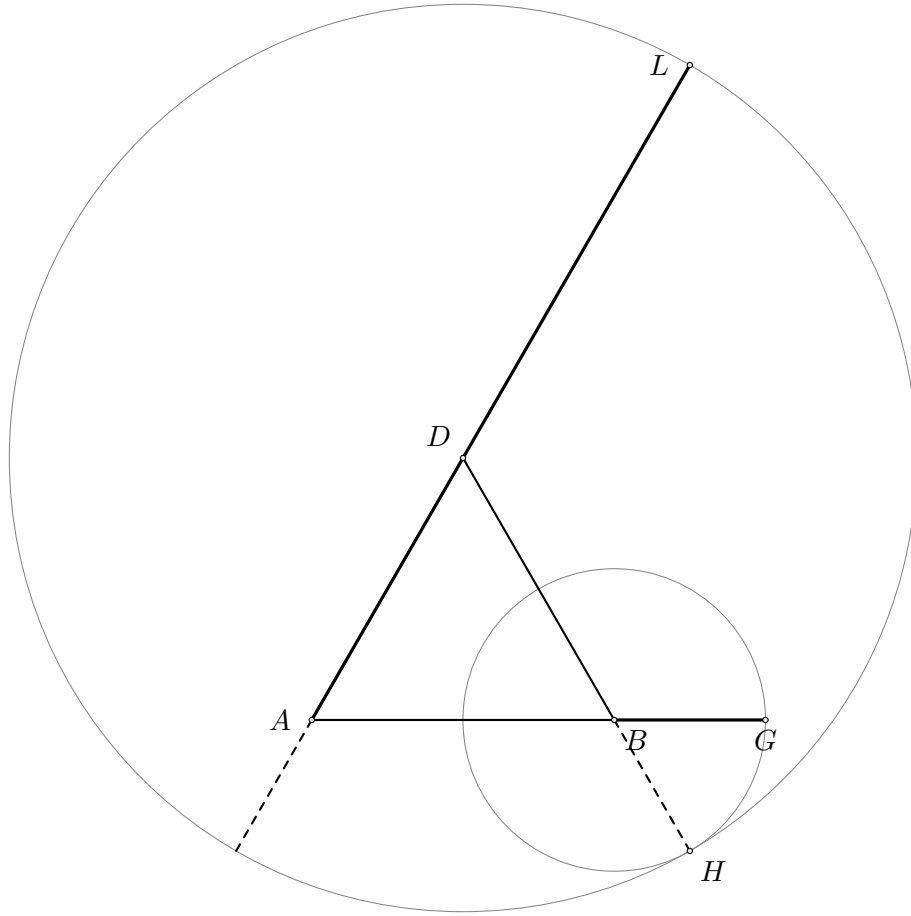


Since  $H$  lies on the circle centered at  $B$  through  $G$ ,

$$BH \cong BG.$$

### Step 6. Draw the circle centered at $D$

Draw the circle centered at  $D$  through  $H$ . Let  $L$  be the intersection of this circle with the line  $DA$  on the side beyond  $A$ .



Since  $L$  lies on the circle centered at  $D$  through  $H$ ,

$$DL \cong DH.$$

The segment  $AL$  is therefore the required copy of the given segment.

## 3 Hilbert Reconstruction

The reconstruction over the Hilbert foundation is considerably shorter.

1. Choose any point  $E \neq A$ .
2. The ray  $AE$  is thereby determined.
3. Apply Book Zero #49 (Layoff).
4. Obtain a point  $D$  on the ray  $AE$  satisfying

$$AD \cong BC.$$

No auxiliary geometric construction is required.

Book Zero #49 provides a point  $D$  on a chosen ray from  $A$  such that

$$AD \cong BC.$$

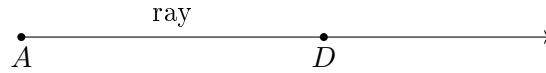


Figure 1: Lay off the segment on a chosen ray.

## 4 Formalization

The formal proof consists of a single application of Book Zero #49.

```
theorem euclid_proposition_2 ...
  := bookZero_49_layoff ...
```

## 5 Discussion

Thus Proposition I.2 illustrates a fundamental architectural difference between Euclid’s geometry and the Hilbert approach.

In Euclid’s *Elements*, the transport of a segment is obtained through a sequence of explicit geometric constructions beginning with Proposition I.1.

In contrast, Hilbert incorporates this operation into the congruence axioms. Book Zero encapsulates it in the Layoff Theorem, making Proposition I.2 an immediate consequence.

## 6 Book Zero #49: Layoff

The proof of Proposition I.2 relies on one of the elementary construction theorems established in Book Zero.

**Book Zero #49 (Layoff).** Let  $A \neq B$  and  $C \neq D$ . Then there exists a point  $X$  on the ray  $AB$  such that

$$AX \cong CD.$$

In other words, every nondegenerate segment can be copied onto an arbitrarily chosen ray while preserving its length.

Input

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Ray AB

A----->

Segment CD

C-----D

|

| Book Zero #49

v

Output

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A-----X----->

AX congruent CD

This theorem provides the fundamental operation of transporting segments. Once it is available, Euclid's Proposition I.2 becomes an immediate consequence.